MLOps-Unit 1		
Experiment Name	**AIM**	**Other information regarding experiment**
1.Telecom Data Preprocessing, Feature engineering and churn Analysis	To preprocess the Telecom customer churn dataset by handling missing values, converting data types, performing feature engineering and analyzing churn patterns to derive business insights	Tools:Python,Pandas,Numpy,Scikitlearn Dataset: Telecom Customer Churn dataset -Handle the missing values -Encoding target variable -Feature engineering -Analyze churn behavior
2. Churn prediction model development	To build and evaluate machine learning models for predicting customer churn using the preprocessed telecom dataset	Tools:Python,Scikit-learn Dataset: Preprocessed Telcom Customer Churn dataset -Define X and Y -Perform Train-Test split -Classifcation models-Logistic regression,Decision Tree,XGBoost -Evaluate model Performance -Compare the Best model and select it
3.Modular ML Pipeline Development for Churn prediction	To design and implement a modular ML Pipeline	Tools: Python Dataset: Telecom Customer Churn dataset Create modular functions:

	integrating preprocessing, feature engineering, model training, and evaluation.	-Data loading -Data preprocessing -Feature Engineering -Model Training -Evaluation Integrate all steps into single pipeline script
4.Version Control of ML Pipeline Using Git	To implement version control for the churn prediction pipeline using Git and GitHub for efficient tracking and collaboration	Tools:Git,Github Dataset: Telcom Customer churn dataset Initialize repository: -Track pipeline code and dataset -Perform commits with meaningful messages -Use branching strategy -Merge changes and push to GitHub
5.CI automation for ML pipeline using GitHub Actions	To automate the churn prediction pipeline using continuous integration with GitHub Actions	Tools: GitHub Actions,Python Dataset:Telecom customer churn dataset Create CI workflow: -Automate Execution of ML pipeline on code push -Validate-Data preprocessing success and model performance Accuracy >=75 -Implement failure conditions if validation does not meet criteria

SVU_Exfedu_MSDS
SVU_Exfedu_MSDS